

BLIND SCOPE-AUDIT BAR for the YM-mass-gap paper (the time-box sprint's

Keeper

2026-07-31

K1061 — Blind scope-audit bar for the YM paper (committed before it lands)

The time-box sprint (to 5 PM) has one dominant risk: scope over-claim — the Clay-problem banner creeping back under deadline pressure. So the checker's half is committed blind now, grounded on the honest scope the corpus already established (K939/K937), so the consolidated paper is ruled against a bar written before it was seen.

The honest scope (K939/K937 — verified, the ground of the bar)

“Derive Yang–Mills” splits three ways: 1. The YM theory/Lagrangian from D_{IV}^5 — CANDIDATE mechanism, not derived. 2. The Clay Millennium problem (interacting QFT *on* R^4 , OS/Wightman, mass gap $\Delta > 0$ there) — NOT derived; the papers say so. The Clay problem is R^4 ; BST's construction is on D_{IV}^5 . 3. The pieces BST does have, at honest tiers: - (A) confinement = no free colored asymptotic states (Schur/Shilov, $\lambda_2 > 0$) — DERIVED (K937/K938). - AF SIGN — DERIVED (K933/K936, one-domain); the 11/3 coefficient is IMPORTED (universal 4D YM, K1052). - Mass-gap VALUE (glueball $c_{2\pi^5 m_e} = 1720$; proton $6\pi^5 m_e = 938$) — IDENTIFIED (matches lattice, not proven). - (B) area-law / linear-potential / R^4 -mass-gap — NOT derived (open) — the large open core.

★★ PASS conditions (all required)

1. Scope explicit: the paper states it constructs YM-with-gap on D_{IV}^5 , and does not claim the R^4 Clay problem.
2. Per-piece tiers held: (A) confinement DERIVED; AF sign DERIVED, 11/3 IMPORTED/marked; gap VALUE IDENTIFIED; (B)/area-law/ R^4 -gap NOT derived (open).
3. W4 open: the YM-8 sector's W4 (cluster decomposition) is named OPEN — Rehren holographic route as the *path*, not a closure. (Scalar sector 5/5 is a separate true statement — do not let it imply the YM-8 sector is 5/5.)
4. Mass-gap senses distinct (K1060): glueball 1720 = the pure-YM/Clay gap (Theorem C); proton 938 = a hadron; $\Lambda_{QCD} \sim 200$ = the running scale — three scales, never conflated.

★ FAIL conditions (any one)

- “solved / proved the Clay / Yang–Mills Millennium problem,” or the “ALL SEVEN MILLENNIUM PROVED” banner reappearing.
- W4 claimed closed without a proof (Rehren asserted as done, not as a route).
- the 11/3 claimed derived (it is imported — Rule 11 / K1052).
- the mass-gap value claimed proven rather than identified.
- (B) area-law / linear confinement / R^4 -mass-gap claimed derived.
- proton (938) and glueball (1720) conflated as “the mass gap.”

★ The honest headline that clears

A substantive construction of Yang–Mills with a mass gap on D_{IV}^5 — confinement and the AF sign derived, the gap value identified, the gap forced by the compactness of the dual Q^5 — with the R^4 obstruction located. It is NOT a solution of the R^4 Clay Millennium Problem: $W4$ (cluster decomposition) and the R^4 transfer remain open. That is a strong, honest paper. Time-box pressure does not relax any line of this bar; an honest-open $W4$ beats a forced closure.

— K1061, Keeper, 2026-07-31. Blind scope bar for the YM paper: D_{IV}^5 construction NOT R^4 Clay; confinement/AF-sign DERIVED (11/3 imported); gap-value IDENTIFIED; (B)/area-law + $W4$ OPEN; glueball $\neq$ proton (K1060). FAIL on any Clay-solved / forced- $W4$ / 11-3-derived / gap-proven / area-law-derived / proton-glueball-conflation. Committed before the consolidated paper lands; re-audit against this. See K939/K937 (honest scope), K1060 (senses), K1052 (11/3 universal), K1040-analog discipline.